

**Description of design data for metals:
Fatigue and Damage Tolerance**

Part: **1.2 – METALLIC MATERIALS**

Chapter: **1260**

Title: **Description of design data for metals:
Fatigue and Damage Tolerance**

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1 INTRODUCTION

The fatigue **and damage tolerance analysis** of a component covers three stages: the crack initiation period which includes micro crack growth, the crack growth period with macro crack growth and the final failure. Each one of these stages is controlled by different material properties.

The aim of this chapter is the detailed description of the material fatigue properties involved in these three stages, for a better understanding of the influence of each parameter in the fatigue life.

Valid sources for material endurance data of metallic materials used in Airbus Defence and Space are: MMPDS (Ref. 1), **HSB material data-sheets (Ref. 6)**, **ESDU data-sheets (Ref. 7)**, **NASGRO (Ref. 5)** and Airbus qualification or Airbus dedicated tests.

Description of design data for metals: Fatigue and Damage Tolerance

2 CRACK INITIATION

The crack initiation is the phase of the total life of the component that covers the nucleation of microscopically small cracks. The crack initiation life then is defined as the total number of cycles before a small crack begins its macroscopic growth (1mm).

The most common method to obtain fatigue data is the axial loading tests. These tests are performed by applying an axial cyclic load, either completely reversed ($\sigma_m = 0$) or those in which the mean stress was varied to create different stress (or strain) ratios ($R = \sigma_{\min} / \sigma_{\max}$ or $\epsilon_{\min} / \epsilon_{\max}$) to consider the mean stress or strain effects. The stress (or strain) is applied with constant amplitude until the specimen fails or a large number of cycles are reached. These tests are carried out for different stress concentrations factors (Kt).

Depending on the stress range that the component supports and whether there is plasticity or not, the better approach for the crack initiation analyses will be either the stress-life approach or the strain-life approach (see chapter 4320: "*Crack initiation life calculation for metallic materials*" for more information).

The stress/strain-life curves must contain at least the following information:

- Temperature: Material fatigue crack initiation properties result affected by temperature; low temperatures generally improve the fatigue properties of the materials, while these are worse at elevated temperatures. Therefore, for those metal alloys exposed to high temperatures, material endurance curves are presented at RT and also at high temperature (see Ref. 1).
- Temper: Most aluminium alloys are tempered by heat treating and/or strain hardening to increase its strength. Since the temper modifies the material properties, it is important to consider it when selecting the material properties for the analysis.
- Product form, i.e. sheet, plate, rolled bar, forgings, **castings or additive manufacturing**. It is important to note that:
 - The thickness indicated in the case of plates corresponds always to the material raw thickness.
 - The thickness indicated in the case of sheets corresponds to the final product. Because of the manufacturing process of sheets, the microstructure changes so that the fatigue behaviour of the material improves when the sheet is thinner. Therefore, it is important to consider the appropriate S-N curve for the crack initiation calculations or being aware in case it is not available that the results obtained could be un/over-conservative.
- Surface conditions: The surface conditions have a significant influence on the fatigue behaviour of the material. Surface treatments modify, both in positive and negative sense, the fatigue resistance. Therefore, surface conditions such as the roughness must be indicated in the S-N curve description.
- Material grain direction: Metallic material fatigue properties are also affected by grain direction. Most fatigue curves are obtained in longitudinal direction (load aligned with material L direction), although for some materials the S-N curves are also defined for long transverse direction (load aligned with material T direction).

Description of design data for metals: Fatigue and Damage Tolerance

- Type and number of specimens, dimensions and stress concentration factor (for stress-life curves): The stresses of the S-N curves, and consequently the K_t , are based on net section.
- Applied loading (i.e. axial load)
- Probability of survival: The fatigue data curves usually represent mean data (50% reliability) but may include an additional factor to produce higher reliability. This is an important aspect to be considered, because the scatter factor used to calculate the threshold is based on this probability of survival.

2.1 STRESS – LIFE APPROACH

The stress-life approach is the method most widely used for the crack initiation analysis and the one with the most accessible material data. The stress-life approach is applicable for high cycle fatigue (components failing after more than $5 \cdot 10^3 - 10^4$ cycles) when the material deformation is primarily elastic.

The basis of the stress – life approach is the use of the material stress – life curves (material endurance data) which describes the relation between the cyclic stress and the number of cycles to failure.

Stress – Life curves

The S-N curves, also called Wöhler curves, are obtained as a result of a number of fatigue tests at different stress levels. For each of the stress level is obtained the number of cycles up to failure (N). Most of the available fatigue data is limited to constant-amplitude axial fatigue data.

- The fatigue life (N) is plotted using a logarithmic scale in the x axis of the diagram.
- The variable (y-axis) is either the stress amplitude (σ_a) or the maximum stress applied (σ_{max}). The stress is plotted on either a linear or a logarithmic scale.

The family of S-N curves obtained are shown schematically for different stress ratios (Figure 2) or for different mean stress (Figure 1).

Some remarks about the S-N diagrams:

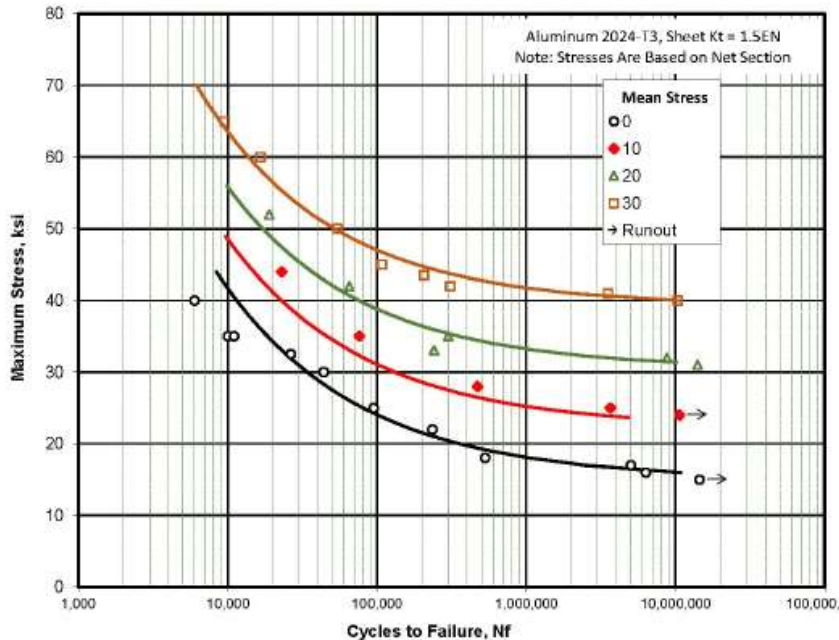
- If the log-log scale is adopted, it leads to a linear relation between $\log(\sigma)$ and $\log(N)$ between $N = 10^2$ and 10^6 cycles, approximately. This linear relation is known as the Basquin equation:

$$\sigma_{max}^k \cdot N = constant$$

- The upper horizontal asymptote corresponds to $\sigma_{max} = \sigma_{ult}$; the specimen would fail in the first cycle if the tensile strength were applied.
- The lower horizontal asymptote is called the fatigue limit (see section 2.1.2).
- Fatigue at high amplitudes and low fatigue lives (up to $N = 10^4$) is called low cycle fatigue, whereas fatigue at lower stress amplitudes and high fatigue lives is called high cycle fatigue.
- This best-fit curve approach is the one used by ADS and implemented in the in-house material database in tabular data form.

Description of design data for metals: Fatigue and Damage Tolerance

Figure 1 shows an S-N diagram from MMPDS (Ref. 1), where the raw test data are plotted for several values of mean stress and a best-fit curve is proposed for each one of them.



Equivalent Stress Equation:

$$\log N_f = 7.5 - 2.13 \log (S_{eq} - 23.7)$$

$$S_{eq} = S_{max} (1-R)^{0.66}$$

Std. Error of Estimate, Log (Life) = 0.30

Standard Deviation, Log (Life) = 0.95

$R^2 = 90\%$

Figure 1. Raw data and best fit S-N curves for notched, Kt = 1.5, 2024-T3 aluminium alloy sheet, longitudinal direction (from Ref. 1)

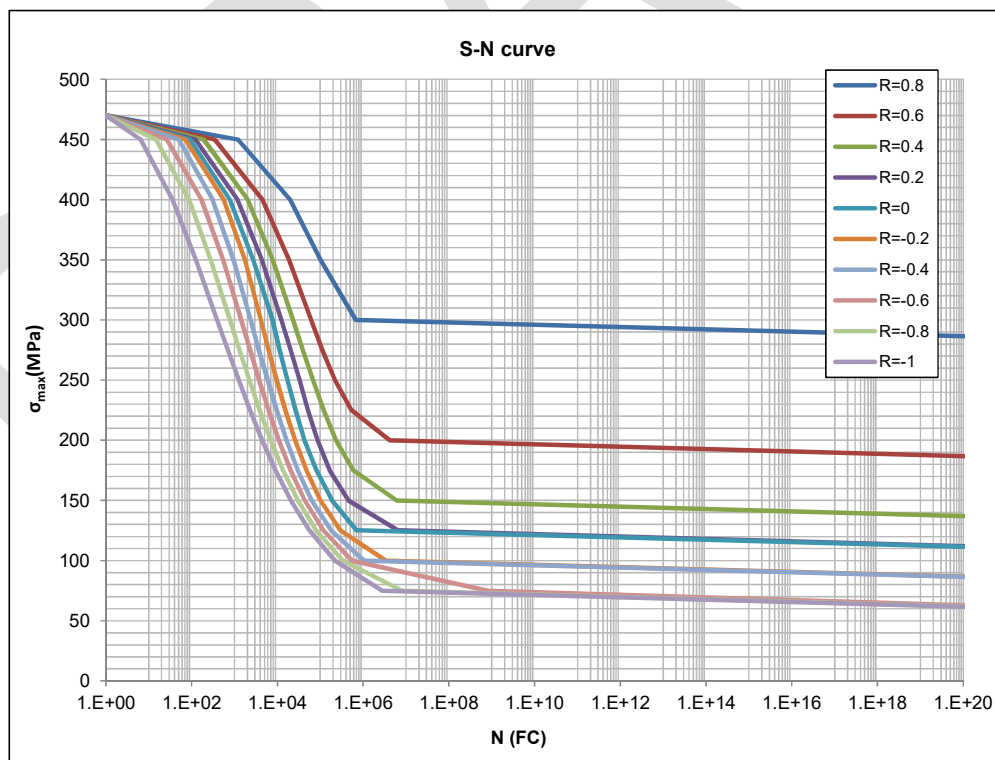


Figure 2. S-N curve for Aluminium 2024-T3 L direction, Kt = 2.5 (fitted data)

Description of design data for metals: Fatigue and Damage Tolerance

Cross plots of the S-N curves obtained for different values of mean stress can be made to obtain a fatigue diagram with lines for constant fatigue life. This is the so called Haigh diagram, where:

- The stress amplitude (σ_a) is plotted in linear scale in the x axis.
- The mean stress (σ_m) is plotted in linear scale in the y axis.

Some remarks about the Haigh diagram:

- All lines for a constant N converge to the same point on the x-axis; $\sigma_m = \sigma_{ult}$ for $\sigma_a = 0$.
- For $N \approx 1$ (static failure) and $\sigma_m = 0$ ($R = -1$), the value of $\sigma_a = \sigma_{ult}$.
- It's interesting to note that the effect of the mean stress is not so important when N is high. That supposes that the stress amplitude has a much larger effect on the fatigue life than the mean stress.

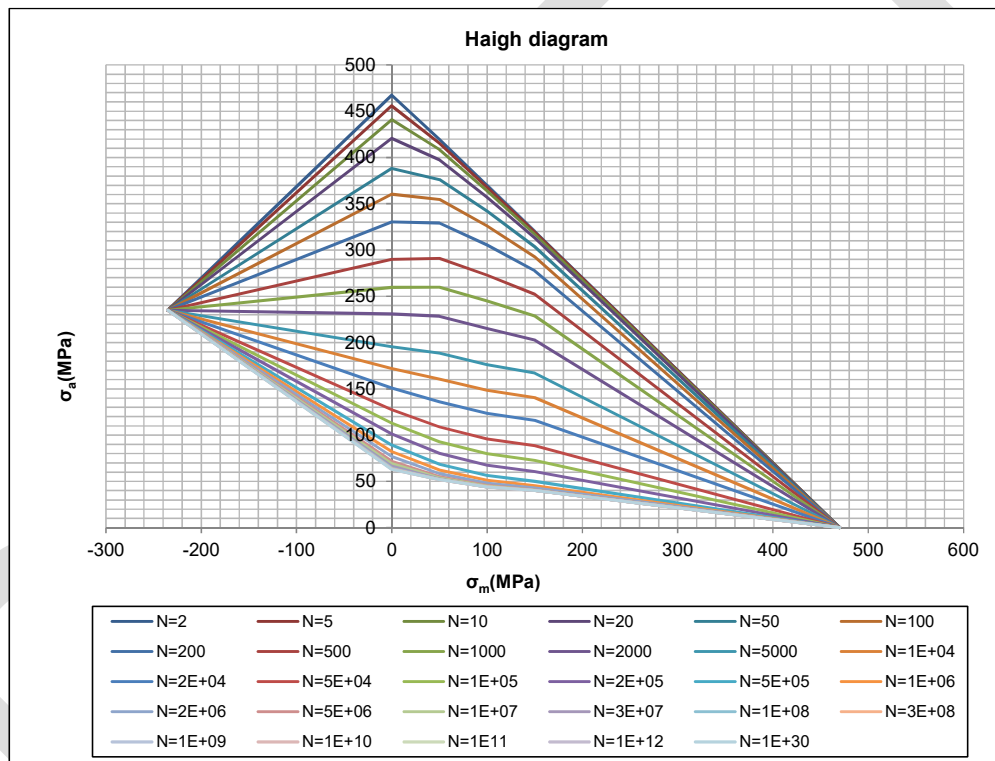


Figure 3. Haigh diagram for Aluminium 2024-T3, $K_t = 2.5$

For more information about how to interpolate/extrapolate for different K_t values, see chapter 4320: "Crack initiation life calculation for metallic materials"

Description of design data for metals: Fatigue and Damage Tolerance

Fatigue Limit

The fatigue (endurance) limit, S_f , represents the lowest stress amplitudes which is capable to nucleate a microcrack that can grow to failure. In other words, it is the maximum stress amplitude that the material is capable to withstand for an infinitely number of cycles with 50% probability of failure.

It is an open discussion, if a true fatigue limit exists. For most ferrous metals, as steels, the S-N curve have well defined a fatigue limit, but for other materials such as for aluminium, the S-N curve continue to drop a slow rate at high number of cycles. This can be seen in Figure 4, where the S-N curves of steel 17-4PH and aluminium 7050-T7451 are compared.

For materials such aluminium some literature recommends to report the fatigue strength rather than the fatigue limit, which is the stress to which the metal can be subjected for a specified number of cycles. Because there is no standard number of cycles, each table of fatigue strengths must specify the number of cycles for which the strengths are reported (see Ref. 8).

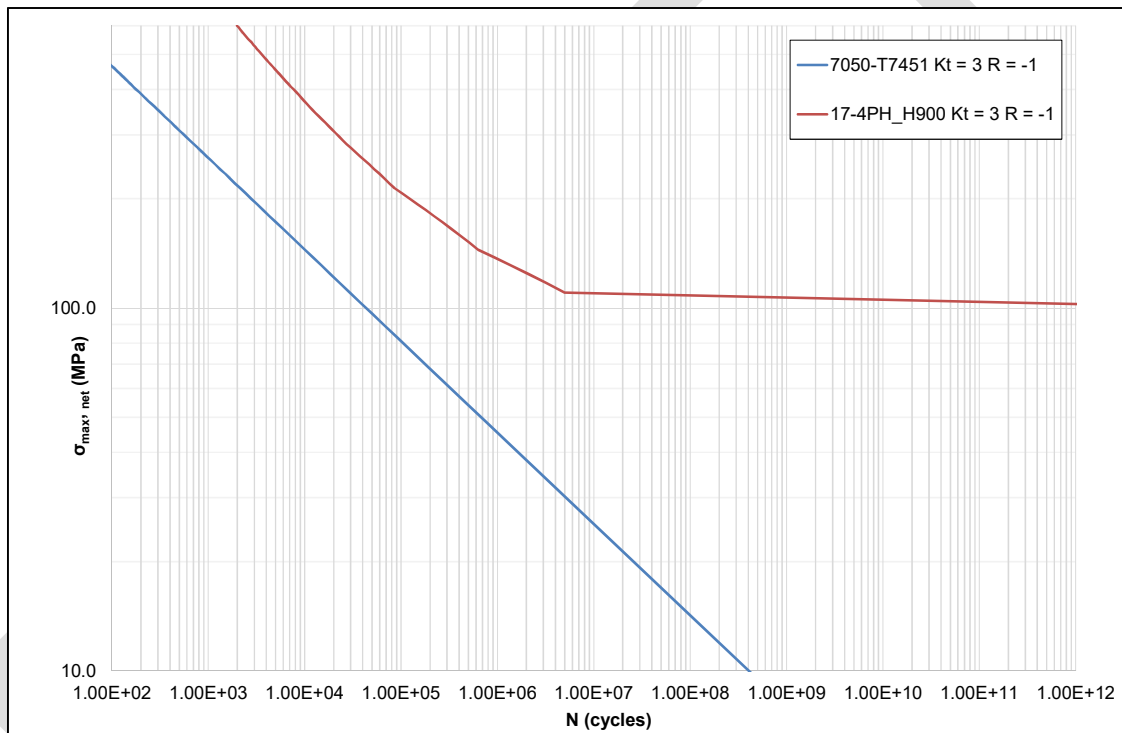


Figure 4. S-N curve for Aluminium 7050T7451 and for Steel 17-4PH

It could be important to point out that for some materials it can establish a proportionality between the fatigue limit for $R = -1$, S_f , and the ultimate stress, S_{ult} : $S_f = \alpha \cdot S_{ult}$. The value of α is around 0.5 for steel, cast iron and Ti-alloys (see Ref. 8). Note that this ratio must be understood just as a first estimation of S_f for unnotched material ($K_t = 1.0$) at $R = -1$.

Fatigue tests to characterize S-N material curves are typically performed in the regime below 10^7 cycles, since very high cycle fatigue tests would need too much time. The highest stress at which a non-failure occurs (runout) is taken as the fatigue limit. Therefore, for those materials that don't exhibit a defined fatigue limit, the challenge is how to calculate the damage in the range of the high cycle fatigue ($N > 10^6$ cycles), where there appear cycles with low stress amplitude and a high frequency.

Description of design data for metals: Fatigue and Damage Tolerance

The original Miner's rule considers that cyclic stresses below the fatigue limit do not accumulate damage in the structure. In order to consider the damage accumulation in this situation, modifications to the traditional Miner's method have been proposed in the literature (see for example Ref. 2 and Ref. 3). These proposals modify the S-N curve slope in the range of the high cycle fatigue in order to take into account the damage of the stresses below the run-outs.

2.2 STRAIN – LIFE APPROACH

The strain-life method is applicable to account for high stress ranges because it considers local plasticity in the material. This method is applicable to low cycle fatigue (the failure appears before 10^4 cycles), where the material deformation is combination of elastic and plastic strains.

The material fatigue properties required for the strain – life approach are the cyclic stress – strain curve and the strain – life curves.

Cyclic stress – strain curve

The material response in a cyclic loaded component is a hysteresis loop of true stress and true strain. Similar to the monotonic relationship, the material cyclic stress-strain curve can be written as:

$$\varepsilon = \frac{\sigma}{E} + \left(\frac{\sigma}{K'}\right)^{1/n'}$$

Where σ and ε are the cyclically stable stress and strain amplitudes respectively, and K' and n' are the cyclic strength coefficient and the cyclic strain hardening exponent respectively.

For most materials the hysteresis loop data can be derived by scaling the cyclic stress-strain curve using Massing's equation (see Ref. 4). It allows obtaining the hysteresis loop by doubling the cyclic stress-strain curve as shown in Figure 5.

The hysteresis loop equation, in terms of ranges, is obtained as:

$$\frac{\Delta\varepsilon}{2} = \frac{\Delta\sigma}{2E} + \left(\frac{\Delta\sigma}{2K'}\right)^{1/n'} \Rightarrow \Delta\varepsilon = \frac{\Delta\sigma}{E} + 2\left(\frac{\Delta\sigma}{2K'}\right)^{1/n'}$$

Description of design data for metals: Fatigue and Damage Tolerance

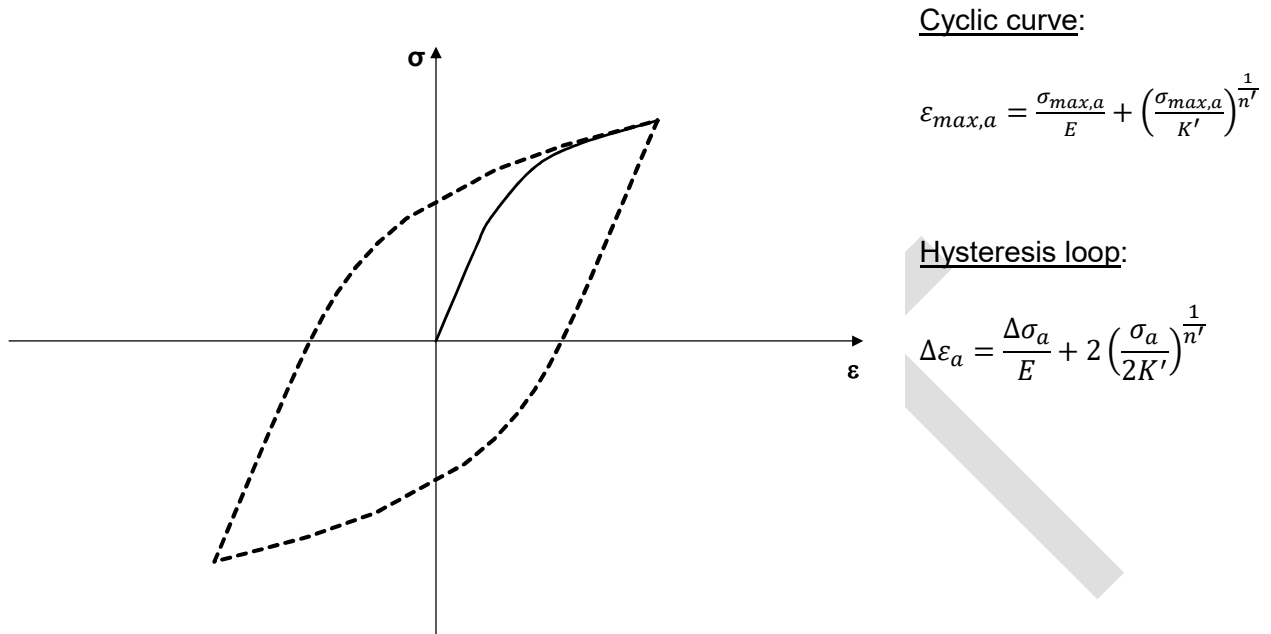


Figure 5. Hysteresis loop and cyclic curve

Figure 6 shows the cyclic stress-strain curve for 7050-T7451 LT direction, taken from Ref. 1:

Figure 6. Cyclic stress-strain curve for 7050-T7451 LT direction (from Ref. 1)

Description of design data for metals: Fatigue and Damage Tolerance

Strain – Life curves

The strain life curve is obtained by cyclic strain-controlled tests where the total strain amplitude (ϵ_a) is obtained as the sum of the elastic (ϵ_{ae}) and the plastic (ϵ_{ap}) strains measurements. From Basquin, Coffin and Manson expressions, the strain-life relationship can be represented as:

$$\frac{\Delta\epsilon}{2} = \frac{\sigma'_f}{E} (2N_f)^b + \epsilon'_f (2N_f)^c$$

The constants in this equation are:

- b is the fatigue strength exponent (Basquin's exponent)
- σ'_f is the fatigue strength coefficient
- c is the fatigue ductility exponent (the Coffin-Manson exponent)
- ϵ'_f is the fatigue ductility coefficient, which is the plastic strain amplitude at $2N_f = 1$
- E is the elastic modulus

Figure 7. Strain-Life curve for 7050-T7451 LT direction (from Ref. 1)

For most materials, the fatigue data of the notch area of a component is represented under null mean strain ($R = -1$). Therefore, a mean strain correction is needed when using this approach when the mean strain is different than zero (see chapter 4320: “Crack initiation life calculation for metallic materials”).

Description of design data for metals: Fatigue and Damage Tolerance

3 CRACK PROPAGATION

The fatigue crack growth is the macroscopic growth of an existing crack under cyclic loads.

The crack growth is a process controlled primarily by the maximum load or stress ratio. Standard testing procedures are documented in ASTM E647 (see Ref. 1). Additional factors that must be specified include:

- **Temperature:** Material fatigue crack growth properties result affected by temperature; high temperatures tends to increase the crack growth rates. This increase of the crack growth rate due to high temperatures is greater for higher R-ratio cycles. In general, material crack growth rate data (da/dN curves) are available at RT and at high temperatures for those alloys exposed to higher temperatures (see Ref. 1).
- **Environment:** The environment is also an important factor in the crack growth. High moisture and saline environments accelerate the crack growth. Material da/dN curves are usually provided at lab air, but for some material data for salt water can be found too.
- **Product form and initial product thickness:** It's important to note that the thickness specified in the da/dN curves is the raw product thickness. Therefore, for the crack propagation analysis, the curve corresponding to the raw product thickness must be chosen. This is particularly important in the case of plates.
- **Crack orientation:** da/dN curves must always specify the crack orientation as an ordered pair of grain direction symbols per ASTM E399. The first letter refers to the grain direction in the load direction. The second letters refers to the grain direction parallel to the fracture plane. Figure 8 shows the six principal fracture path directions for flat sections and cylindrical shapes:

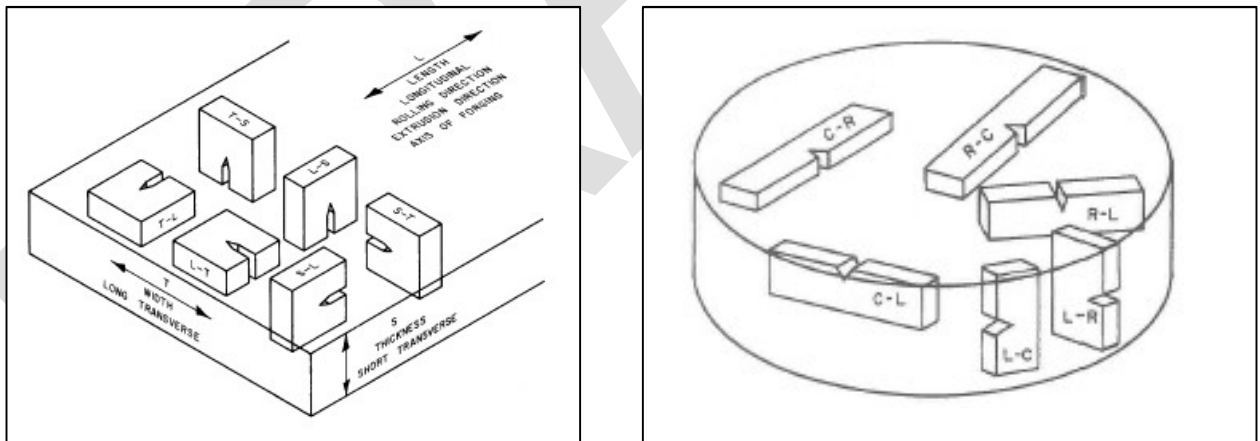


Figure 8. Typical principal fracture path directions (from ASTM E399)

Before talking about the da/dN curves, it has been considered important to summarize some general definitions, although they will be treated more deeply in future chapters of this SMM:

Description of design data for metals: Fatigue and Damage Tolerance

Stress intensity factor (SIF):

The stress intensity factor, also abbreviated as SIF or K, describes the severity of the stress distribution around the crack tip. The SIF, for a finite width sheet, with a single through crack of total length $2 \cdot a$ and submitted to a constant stress level σ in the far field is defined as:

$$K = \sigma \cdot \sqrt{\pi \cdot a}$$

In a general case, the above expression is modified with a correction factor β , called normalised stress intensity factor, to consider different geometrical effects, load application and interaction with the rest of the cracks:

$$K = \beta \cdot \sigma \cdot \sqrt{\pi \cdot a}$$

The β factors depend on the geometry and the load case and are calculated for each crack tip and for each crack length considered in the crack propagation analysis.

The SIF for different geometries and loading conditions can be found in literature. The results are often presented as values of the correction factor β in graphs that show how β depends on geometry ratios or loading conditions.

Crack growth curve

Constant-amplitude data consist of crack length measurements at corresponding loading cycles. Such data are presented as crack growth curves. Figure 9 shows a typical crack growth data, where the crack length is plotted versus the corresponding number of cycles N at which the crack is measured or calculated.

The crack growth rate, da/dN , is defined as the crack extension that the crack grows in a cycle and it can be calculated as the slope of the crack growth curve at each crack length.

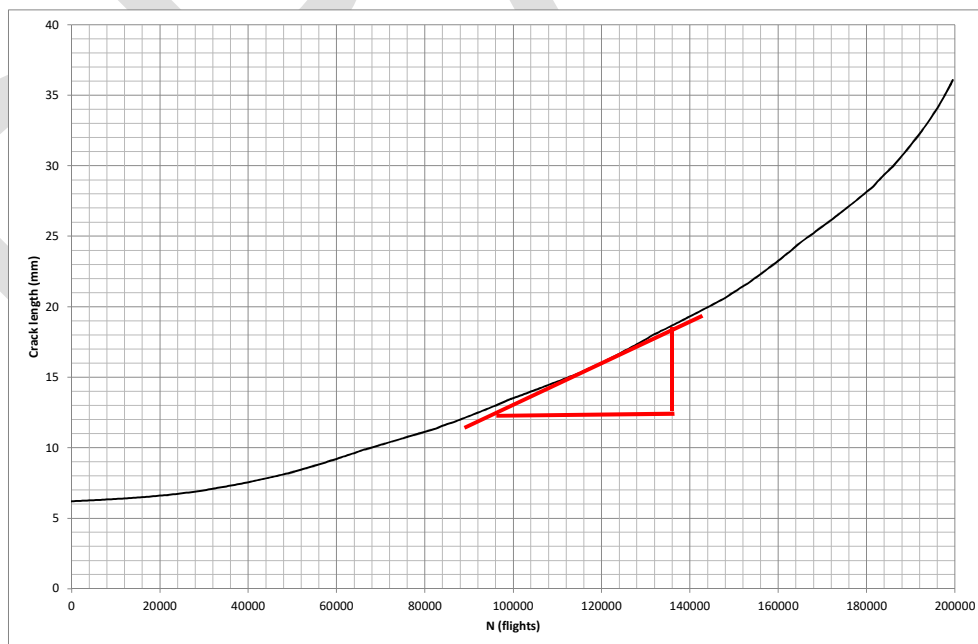


Figure 9. Crack growth data

Description of design data for metals: Fatigue and Damage Tolerance

3.1 da/dN CURVES

Fatigue crack-propagation data presented in the form of crack growth curve (a vs N) is not useful for design purposes since a variety of stress levels, stress ratios and initial crack conditions are encountered. Therefore, the fatigue-crack-propagation damage behaviour of materials is modelled as rate process and formulated as a dependent variable based on the slope of the crack growth curve (da/dN).

Crack growth rate, da/dN, is presented as a function of the stress intensity factor range, ΔK , and the stress ratio:

$$\frac{da}{dN} = f(\Delta K, R)$$

This function is usually presented as logarithmic plots of da/dN versus ΔK .

Figure 10 shows a typical plot of da/dN data versus ΔK (log-log scale). This curve is divided in three regions:

- Region I: When evaluating the crack growth, it is observed that there is a value of the stress intensity factor range (ΔK_{th}) below which the growth is negligible. The stress intensity factor range threshold is not a single material constant because it depends on the stress ratio R: ΔK_{th} decreases when the stress ratio (R) increases.

It's important to note that the value of ΔK_{th} is not associated with the fatigue limit (see section 2.1.2). The threshold value ΔK_{th} is concerned with existing cracks that have grown to a macroscopic size at a ΔK level above ΔK_{th} , but when ΔK is decreased, the crack growth slows down until it can be assumed that no further growth occurs if $\Delta K < \Delta K_{th}$.

- Region II: It is the largest and most important part of the curve.
- Region III: The region III is characterized by rapid, unstable crack growth. The crack propagation rate grows up as the maximum stress intensity factor approximates to the fracture toughness of the material.

The transition between region II and region III depends on the material yield stress, the stress intensity factor and the stress ratio

There are different mathematical approaches developed to model the da/dN behaviour:

- For Region I the crack growth rate can be predicted by using the following equation:

$$\frac{da}{dN} = C(\Delta K - \Delta K_{th})^p$$

The crack propagation threshold has been calculated in Airbus legacy programs as:

$$\Delta K_{th} = K_{tho} - m \cdot R$$

Where the constants C, K_{tho} , m and p depend on the material type.

Description of design data for metals: Fatigue and Damage Tolerance

- Paris and Erdogan equation: To predict the fatigue crack growth in the region II of the curve

$$\frac{da}{dN} = C(\Delta K)^n$$

- Elber: It's a modification of the Paris equation, used to model the region II of the curve. This model takes into account the stress ratio effect, which is one of the most important limitations of the Paris equation:

$$\frac{da}{dN} = C \cdot [(A + B \cdot R) \cdot \Delta K]^n$$

Where C, A, B and n are material constants.

- Forman developed an equation to model the material behaviour in the crack growth regions II and III. It considers the increased growth rate when approaching to the fracture toughness. Forman's equation is described as:

$$\frac{da}{dN} = \frac{C(\Delta K)^n}{(1-R) \cdot K_c - \Delta K} = \frac{C(\Delta K)^n}{(1-R) \cdot (K_c - K_{max})}$$

All the material constants for each one of these equations can be found in literature or calculated by performing tests (ASTM E647).

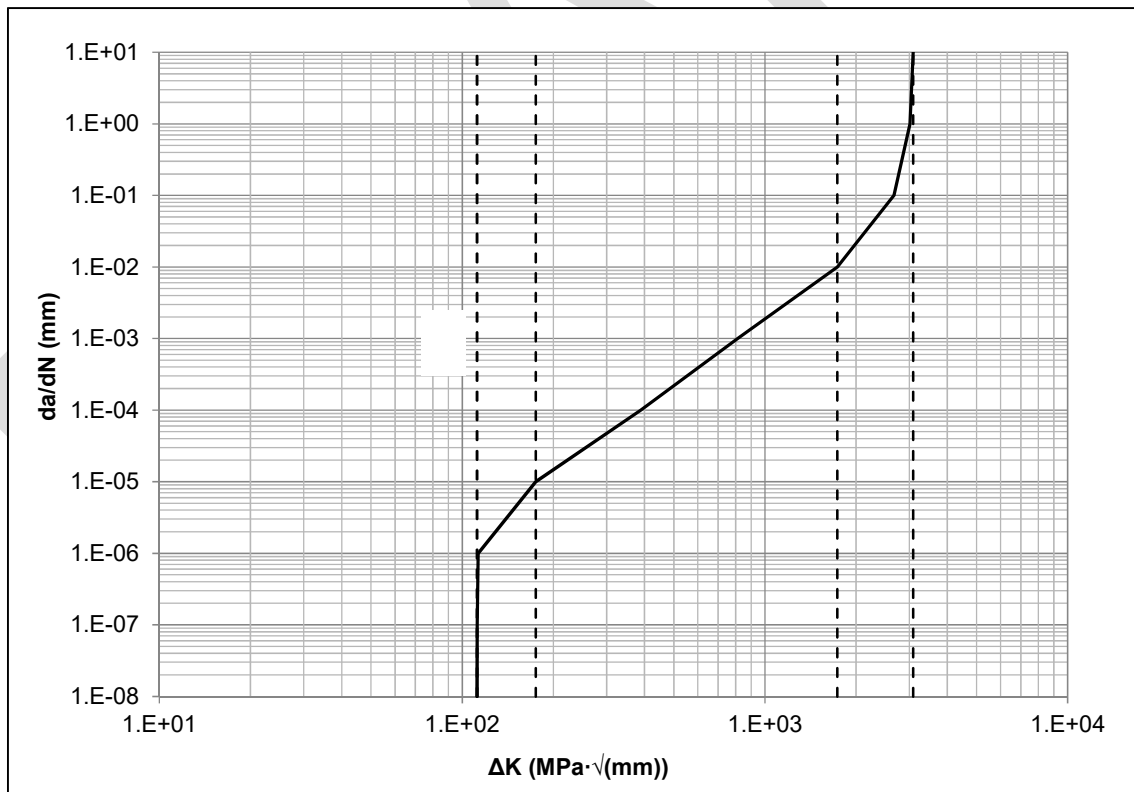


Figure 10. da/dN curve for 2024-T3 L-T

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4 FRACTURE TOUGHNESS

As with fatigue crack growth properties, fracture toughness depend in the same manner on the product form, environment and crack orientation.

The effect of temperature on toughness depends on the alloy base. The fracture toughness usually decreases when the temperature is decreased. However, the temperature effecting data in references are very limited.

4.1 K_{Ic} , K_{Ic} AND K_{app}

The material fracture toughness, abbreviated as K_{Ic} , is the critical value of the stress intensity factor that leads to fracture.

The fracture toughness is a material property that depends mainly on the stress state, i.e., plane-stress (thin thickness) or plane-strain (thick thickness). When the thickness increases the fracture toughness tends to a stable lowest value (K_{Ic}).

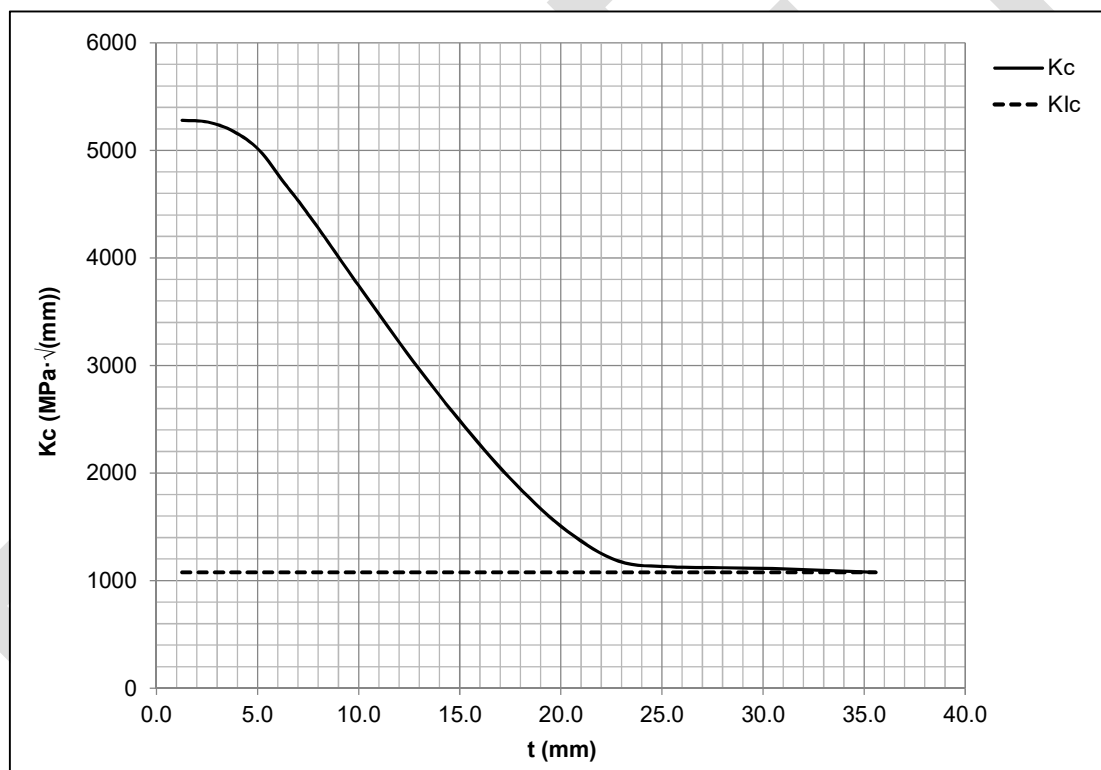


Figure 11. K_c curve for 2024-T3 Sheet LT direction

Description of design data for metals: Fatigue and Damage Tolerance

Fracture toughness data is used for calculating the critical crack length when using the residual strength criterion and to calculate the crack instability. Depending on the stress state it is obtained as:

- Plane strain: K_{Ic} must be used.
- Plane stress: The curve K_{app} must be used to obtain the corresponding fracture toughness to the component thickness. The difference between K_{app} and K_c is shown in Figure 12. K_c (point C) is calculated considering the maximum stress level and the critical crack length, while K_{app} is conservatively calculated as a function of the maximum stress and the initial flaw size (point A).

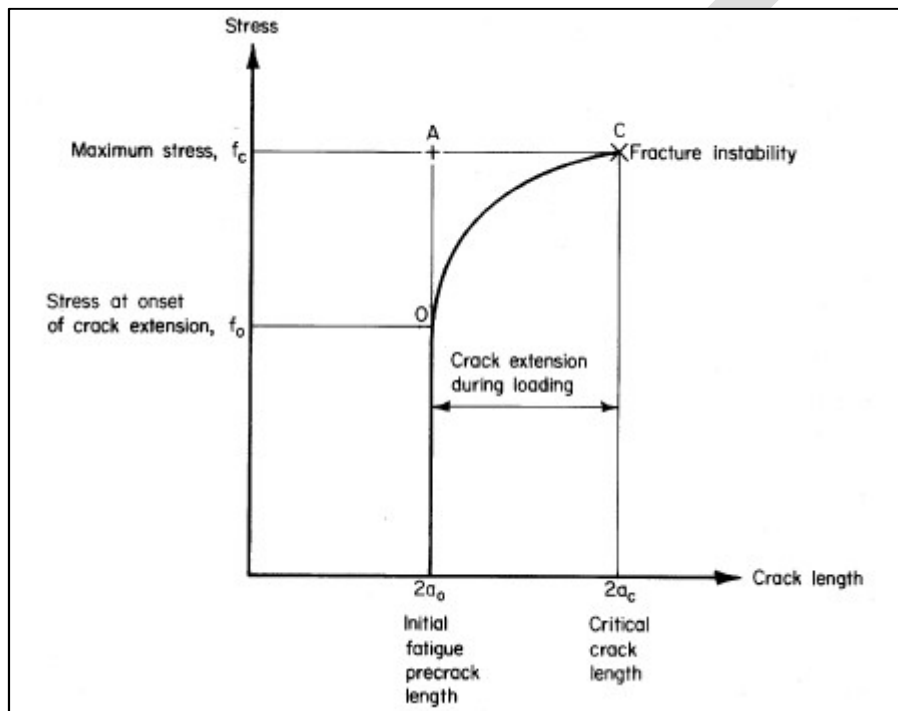


Figure 12. Applied stress vs Crack length (from Ref. 1)

It is recommended to use K_{app} curve instead of K_c . However, and due to the lack of information for some of the most used materials, K_c curve has been used in legacy programs.

For those materials for which the apparent fracture toughness curve is not available, it can be calculated according to the expression proposed by NASGRO:

$$t_0 = 2.5 \cdot \left(\frac{K_{Ic}}{\sigma_{yield}} \right)$$

$$K_{c0} = K_{Ic} \cdot \left[1 + B_K \cdot e^{-\left(A_K \frac{t}{t_0} \right)^2} \right]$$

Where; K_{Ic} is the plain strain fracture toughness, σ_{yield} is the material yield stress and A_K and B_K are material constants from NASGRO.

Description of design data for metals: Fatigue and Damage Tolerance

4.2 K_R CURVES

The material crack resistance curve (R-Curve) is the representation of K_R vs Δa , where K_R represents the stress intensity factor required to produce stable crack growth previous to the complete catastrophic failure at K_c .

The R-curve is obtained by tensions tests of centred cracks with length equal to $2a$.

Figure 13 illustrates the stress intensity factor (K_R) versus the effective crack extension for the half crack length (one tip only) for material 2024-T3:

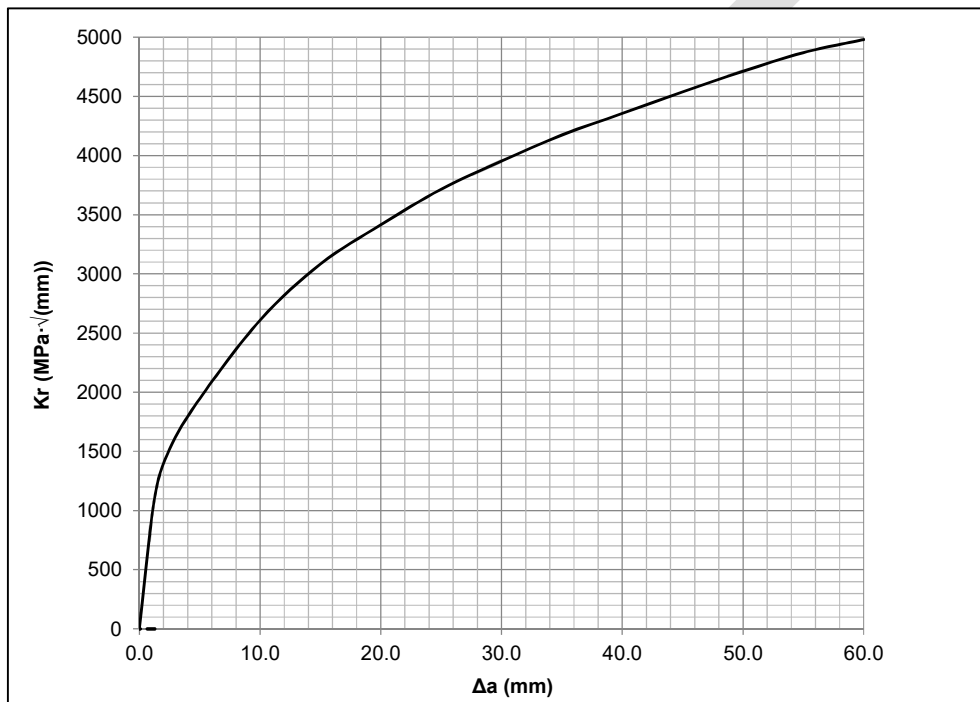


Figure 13. K_R curve for 2024-T3 Sheet LT direction

The R-curve is used to calculate the critical crack length by comparison with the stress intensity factor at limit load.

Description of design data for metals: Fatigue and Damage Tolerance

5 STRESS CORROSION - K_{Isc}

The stress corrosion phenomena is the crack initiation and growth under a sustained load or residual stress. The material property that indicates how sensitive the material is to stress corrosion is K_{Isc} , which is defined as a percentage of K_{Ic} . For example, some titanium alloys exhibit $K_{Isc} > 0.8 \cdot K_{Ic}$.

The stress corrosion resistance depends on the material type, temper, product grain direction and magnitude of the stress. Then, for example, the T74 and T76 tempers of 7010, 7075, 7475, 7049, 7149 and 7050 provide an intermediate degree of resistance to stress-corrosion cracking, superior to that of the T6 temper, but not as good as the T73 temper of 7075 (Ref. 1).

To assist in the selection of materials Ref. 1 presents letter ratings indicating the relative resistance to stress corrosion cracking of various mill product forms of the most common used Aluminium.

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Description of design data for metals: Fatigue and Damage Tolerance

6 DATASHEET TEMPLATE FOR A METALLIC MATERIAL: FATIGUE PROPERTIES

MATERIAL IDENTIFICATION

Name/designation	
Version	
Description (Temper and Form)	
Specifications	
Equivalences	
References	
Units	International System of Units (SI): subset of derived units (N,mm)

S-N curves

Material Parameters	Symbol	Units	Value
S-N curve number	--	N/mm ² - cycles	Airbus DS Materials Database reference number
Thickness	t	mm	
Direction	--	--	L, LT, ST
Kt	Kt	--	
Format	--	--	$\sigma_{\max}$ -R, $\sigma_{\max}$ - σ_m , σ_a - σ_m , $\sigma_{\max}$ - $\sigma_{\min}$, N=const
Temperature	T _{ref}	°C	
Reference / Source			

ϵ -N curves

Material Parameters	Symbol	Units	Value
ϵ -N curve number	--	Strains – cycles	Airbus DS Materials Database reference number
Thickness	t	mm	
Direction	--	--	L, LT, ST
Kt	--	--	
R	--	--	
Temperature	T _{ref}	°C	
Reference / Source			

Description of design data for metals: Fatigue and Damage Tolerance

Cyclic curves

Material Parameters	Symbol	Units	Value
Cyclic curve number	--	N/mm ² - Strains	Airbus DS Materials Database reference number
Thickness	t	mm	
Direction	--	--	L, LT, ST
Young Modulus	E	MPa	
Yield tension stress	F _{ty}	MPa	
Ultimate tension stress	F _{tu}	MPa	
Proportional Limit	L _p	MPa	
Temperature	T _{ref}	°C	
Reference / Source			

da/dN curves

Material Parameters	Symbol	Units	Value
da/dN curve number	--	mm – N/mm ² *√mm	Airbus DS Materials Database reference number
Thickness	t	mm	
Direction	--	--	L-T, T-L, S-L, L-S, T-S, S,T
Format	--	--	Database, Forman equation, Elber equation
Temperature	T _{ref}	°C	
Reference / Source			

Kc curves

From ADS materials database:

Material Parameters	Symbol	Units	Value
Kc curve number	--	N/mm ² *√mm – mm	Airbus DS Materials Database reference number
Direction	--	--	L-T, T-L, S-L, L-S, T-S, S,T
Plain strain fracture toughness	K _{Ic}	N/mm ² *√mm	
Temperature	T _{ref}	°C	
Reference / Source			

Description of design data for metals: Fatigue and Damage Tolerance

From NASGRO material properties database:

Material Parameters	Symbol	Units	Value
Direction	--	--	L-T, T-L, S-L, L-S, T-S, S,T
Plain strain fracture toughness	K_{Ic}	$N/mm^{2*\sqrt{mm}}$	
Ak	--	--	Coefficient from NASGRO material properties database
Bk	--	--	Coefficient from NASGRO material properties database
Temperature	T_{ref}	°C	
Reference / Source	NASGRO material properties database		

Kapp curves

Material Parameters	Symbol	Units	Value
Kapp curve number	--	$N/mm^{2*\sqrt{mm}} - mm$	Airbus DS Materials Database reference number
Direction	--	--	L-T, T-L, S-L, L-S, T-S, S,T
Plain strain apparent fracture toughness	K_{Ic}	$N/mm^{2*\sqrt{mm}}$	
Temperature	T_{ref}	°C	
Reference / Source			

Kr curves

Material Parameters	Symbol	Units	Value
Kr curve number	--	$N/mm^{2*\sqrt{mm}} - mm$	Airbus DS Materials Database reference number
Thickness	t	mm	
Direction	--	--	L-T, T-L, S-L, L-S, T-S, S,T
Temperature	T_{ref}	°C	
Reference / Source			

Description of design data for metals: Fatigue and Damage Tolerance

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
a	mm	Crack length
ADS	---	Airbus Defence and Space
b	---	Fatigue strength exponent (Basquin's exponent)
c	---	Fatigue ductility exponent (Coffin-Manson exponent)
E	MPa	Young modulus
K, SIF	MPa√mm	Stress intensity factor
K'	---	Cyclic strength coefficient
K _c	MPa√mm	Fracture toughness
K _{Ic}	MPa√mm	Fracture toughness for plain strain
K _{app}	MPa√mm	Apparent fracture toughness
K _t	---	Stress concentration factor
L	---	Longitudinal direction
N	FC	Number of cycles to failure
n'	---	Cyclic strain hardening exponent
R	---	Stress ratio
RT	---	Room temperature
S, ST	---	Short transverse direction
t	mm	Thickness
T, LT	---	Long transverse direction
β	---	Normalized stress intensity factor
ε	---	Strain
ε _a	---	Strain amplitude
σ _a	MPa	Stress amplitude
σ _m	MPa	Mean stress
σ _{max}	MPa	Maximum stress
σ _{ult}	MPa	Ultimate stress
σ _{yield}	MPa	Yield stress

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REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	MMPDS	13	Metallic Materials Properties Development and Standardization
Ref. 2	Haibach E.		Modifizierte lineare Schadensakkumulationshypothese zur Berücksichtigung der Dauerfestigkeit mit fortschreitender Schädigung.
Ref. 3	X022RP1816165	1.0	Very High Cycle Fatigue (VHCF) method for metallic materials
Ref. 4	G.Massing	1926	Eigenspannungen und Verfestigung beim Messing
Ref. 5	NASGRO	3.0	Fatigue crack growth computer program „NASGRO” Version 3.0 – Reference Manual
Ref. 6	HSB		Handbuch Struktur Berechnung
Ref. 7	ESDU		HIS ESDU Series
Ref. 8	--	1986	Atlas of fatigue curves

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Software programs referenced in this chapter

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.1 08/02/2021	M.Carmen Blanquer Darío Fernández	First issue	All pages

Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Analysis (TEPAA)	Javier Gómez-Escalonilla 08/02/2021	Holger Hicketier 08/02/2021
Stress Methods (TEPAP-TL4)	Darío R. Fernández Villalba 08/02/2021	-

Approval signatures table for last revision of this chapter